

SupplyNerve AI

AI-Powered Supply Chain Intelligence & Decision Platform

Detect Risks. Prioritize Critical Issues. Enable Proactive Decisions.

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Problem Statement

Modern supply chains face demand variability, supplier delays, inventory shortages, and sourcing constraints. Procurement and planning teams spend significant time analyzing fragmented data across multiple systems, causing critical risks to be identified too late and resulting in operational disruptions, higher costs, and reduced productivity.



Key Challenges



Demand Variability

Sudden forecast changes and unexpected demand fluctuations.



Supplier Delays

Late deliveries and commitment failures



Fragmented Data

Information scattered across ERP System, SpreadSheet, emails



Inventory Shortages

Material shortages impacting production schedules.



Sourcing Constraints

Long lead times and limited supplier capacity.



Late Risk Detection

Critical issues identified only after operational impact occurs

Business Impact



Higher Costs



Production Disruptions



Reduced Efficiency



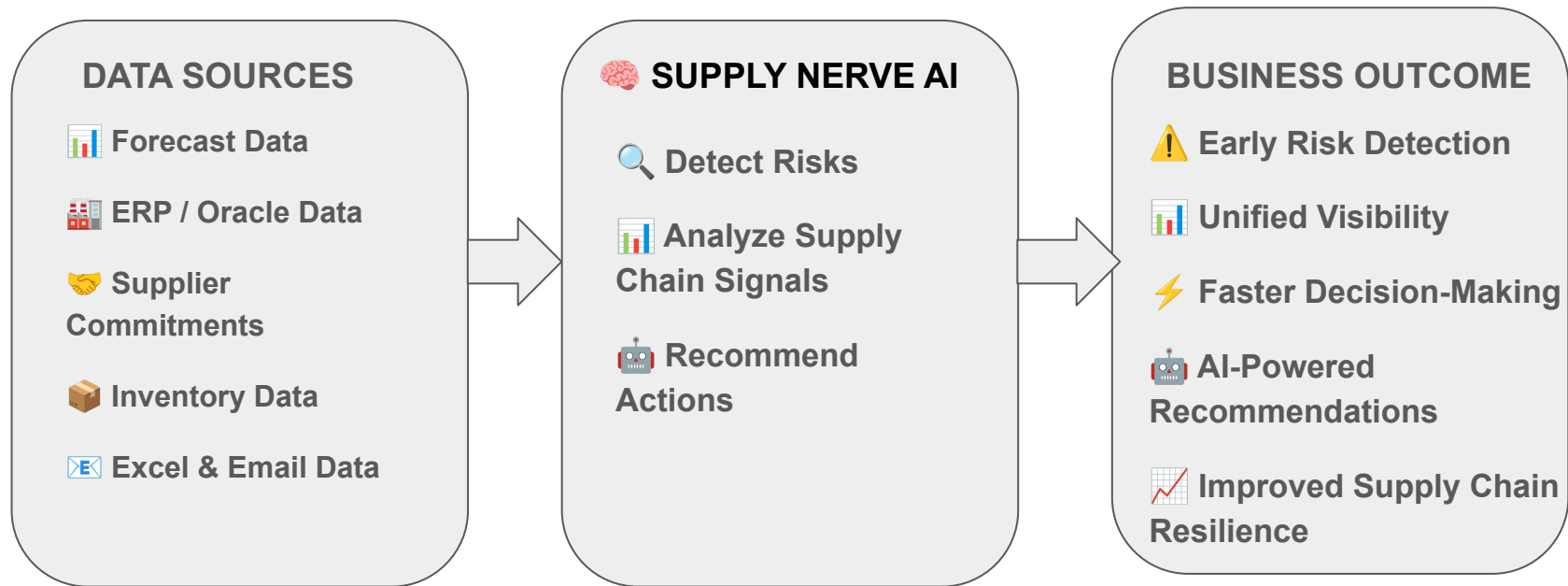
Reduced Productivity



Lower Customer Satisfaction

Solution Overview: SupplyNerve AI

AI-Powered Supply Chain Risk Intelligence Platform



SupplyNerve AI analyzes supply chain signals to detect risks early, prioritize critical issues, and recommend proactive actions before disruptions occur.

Core Capabilities of SupplyNerve AI

AI-powered capabilities designed to improve supply chain visibility, risk detection, and decision-making



Forecast Verification Agent

Detects

- Forecast increases and decreases
- Pull-ins and push-outs
- Demand fluctuations



Benefit

Early visibility into demand changes.



Supplier Commitment Tracker

Monitors

- Supplier responses
- Commitment status
- Missing or delayed confirmations



Benefit

Improved supplier accountability.



Risk Prioritization Engine

Identifies

- High-risk parts
- Supply shortages
- Supplier delays
- Critical exceptions



Benefit

Focus on the most urgent issues first.



AI-Powered Recommendations

Suggests

- Supplier follow-up actions
- Capacity validation
- Inventory review
- Escalation recommendations



Benefit

Faster and more informed decisions.



Automated Alerts & Visibility

Provides

- Risk dashboard
- Exception monitoring
- Alert notifications
- Action tracking



Benefit

Enhanced supply chain visibility.



SupplyNerve AI combines risk detection, supplier monitoring, and AI-driven recommendations to help SCM Team act before disruptions occur

How SupplyNerve AI Works



 Supply Nerve transform complex Supply chain data into clear insights, helping SCM team act before Disruption occur

SupplyNerve AI Prototype Demonstration

AI-powered prototype demonstrating automated risk detection, prioritization, and decision support.

SupplyNerve AI

AI-Powered Supply Chain Risk Intelligence Platform

Upload Excel File

Upload

200MB per file • XLSX

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Excel Upload

AI Recommended Actions

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AI Recommendations

	Part No	Risk	AI Recommendation
0	P1001	High Risk	I recommend escalating the supplier relationship with the current supplier to explore .
1	P1002	Low Risk	In light of the forecast change percentage of 10.00, I recommend that we proceed with
2	P1003	High Risk	I recommend a procurement strategy of "Inventory Holding" to mitigate the potential
3	P1004	Low Risk	I would recommend a 2-year contract with a fixed price of \$150,000 to the supplier, giv
4	P1005	High Risk	Based on the provided information, a short procurement recommendation would be t
5	P1006	Low Risk	I recommend procurement a lot of P1006 parts, as the forecast change percentage of 3
6	P1007	High Risk	I recommend procuring the P1007 component from a reliable supplier, but I will condu
7	P1008	Low Risk	I would recommend procuring the Part P1008 from a reliable supplier with a response
8	P1009	High Risk	I would recommend exploring alternative suppliers to mitigate the risk of a potential i
9	P1010	Low Risk	Given the forecast change percentage of 2.86 and the supplier response of "Yes", I reco

Risk Analysis

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Risk Analysis Output

	Part No	Prev Forecast	Current Forecast	Supplier Response	Spend	Lead Time	Forecast Change %
8	P1009	250	500	No	60000	14	100
9	P1010	350	360	Yes	15000	4	2.8571
10	P1011	220	400	No	55000	11	81.8182
11	P1012	180	175	Yes	12000	5	-2.7778
12	P1013	600	1000	No	95000	18	66.6667
13	P1014	320	340	Yes	22000	7	6.25
14	P1015	140	280	No	47000	12	100
15	P1016	260	270	Yes	18000	6	3.8462
16	P1017	450	800	No	85000	17	77.7778
17	P1018	210	215	Yes	14000	5	2.381

MVP demonstrates automated risk detection, prioritization, and AI-assisted decision support using real-world supply chain scenarios

Business Impact & Expected Benefits



Reduced Manual Effort

- Automated risk monitoring
- Less spreadsheet analysis



More time for strategic work



Faster Risk Detection

- Early identification of risks
- Prevents late escalations



Proactive issue resolution



Improved Visibility

- Unified supply chain view
- Centralized monitoring



Better decision-making



AI-Assisted Decisions

- Intelligent recommendations
- Prioritized actions



Faster response times



Better Supplier Management

- Tracks commitments
- Improves accountability



Stronger supplier collaboration



Improved Forecast Monitoring

- Detects demand fluctuations
- Highlights critical changes



Early demand visibility



Reduced Operational Costs

- Fewer shortages
- Reduced disruptions



Lower business costs



Supply Chain Resilience

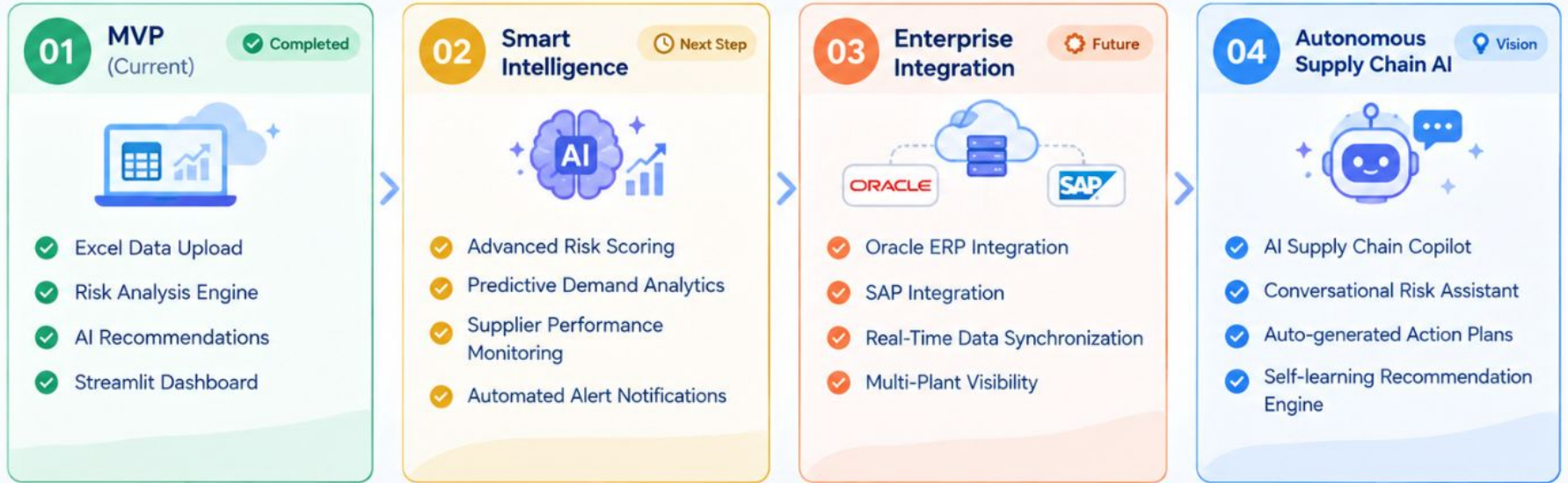
- Better preparedness
- Risk mitigation support



Improved business continuity

SupplyNerve AI empowers procurement and planning teams with early risk visibility, AI-powered recommendations, and proactive decision-making.

Future Roadmap

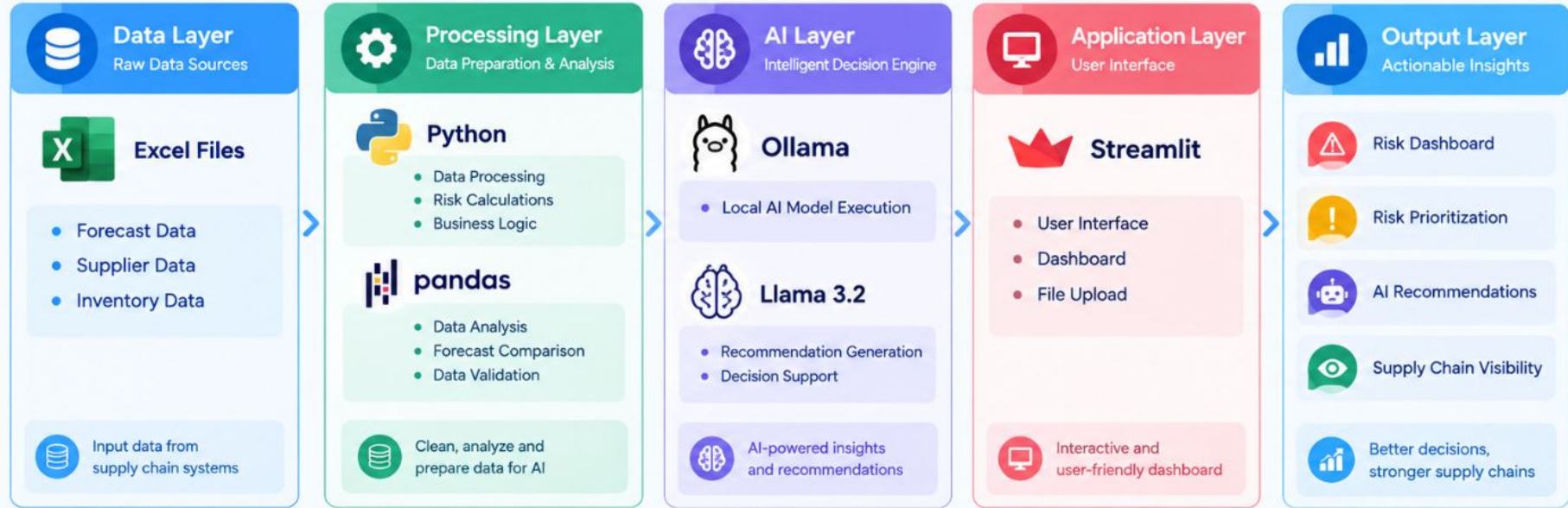


VISION

Transform supply chain management from reactive issue handling to proactive and AI-driven decision intelligence.



Technology Stack



ARCHITECTURE SUMMARY

SupplyNerve AI combines Python analytics, AI-powered recommendations, and an interactive Streamlit dashboard to enable proactive supply chain decision-making.

Thank you!!